

Part B – Weekly Capstone Assignment Template

This template is used for **each of the 14 Part B capstone submissions** throughout the semester. The structure remains the same every week so that you can focus on *thinking and judgment*, not guessing expectations.

Your goal is not to achieve the best performance, but to **reason carefully about how this week's machine learning technique applies (or does not apply) to your project**.

1. Project Context (Brief)

Provide the same short context each week so graders can orient quickly.

- **Project Title: MLB Home Runs**
 - ****Data Modality: Tabular **** (tabular, image, audio, text, other)
 - **Task Type: Regression** (regression, classification, clustering, sequence modeling, etc.)
 - **One-Sentence Goal: Predict how many home runs MLB players will hit next season.** What are you trying to predict, detect, or understand?
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2. This Week's Technique and Its Assumptions

- **Technique / Model Family Covered This Week: Unsupervised Learning**
- ****Key Assumptions of This Technique:**
 - Groups exist within the data that are visible on graphed.** (1–2 bullets)

****Fit Assessment (required): ****

I expect this technique to be a **partial** fit for my project because: I already have a general idea of the structure of the data, rather than needing to find unknown structures in the data. It's a project that naturally lends to supervised more. (2–4 sentences)

3. Representation or Proxy Used

Describe how your data was represented so that this week's technique could be applied.

Examples include:

- Hand-engineered features
- Summary statistics
- Frozen embeddings
- Dimensionality reduction
- A proxy task

- **Representation or Proxy Chosen: Maintain at Hand-engineered features, adding in SLG.**
 - ****Why this representation was reasonable for this week: Same as previous, but now added the new column of slugging. ****
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4. What Was Attempted

Be concrete and scoped. Do not list everything you *could* have done.

- What you implemented this week: Added in SLG, and tried PCA.
- What you intentionally did *not* attempt and why: Did not attempt any other data, although it was tempting. I also almost was going to see the difference running it with SLG or OPS would do, but I wanted to keep it simple.
- Any constraints encountered (data, labels, compute, time): I wanted to maybe add ISO, but my data didn't include that as a column.

5. Results or Observations

You may include metrics **if applicable**, but qualitative observations are also valid.

Examples:

- Evaluation metrics
- Training behavior or convergence issues
- Error patterns
- Unexpected behaviors

The main difference this time was from adding the SLG data in, which gave us a jump on F1 from 0.7 to 0.75 which is a new high.

The graph I got was not particularly interesting for a binary classification, just very split down the middle for "power" vs not hitters.

6. Interpretation and Judgment

This section matters most.

Reflect on:

- Why the method behaved as it did
- Which assumptions held or failed
- What this reveals about your data or problem framing

(1–2 thoughtful paragraphs)

The results of the clustering aligned in a way that makes sense with the data and threshold I had picked for supervised. However it found that mark naturally, so 20 seems like a good point to have picked. I think that was due to the addition of SLG possibly, that a distinction emerged once that data was added.

7. Forward-Looking Adjustment

Answer **one** of the following:

- What will you keep, change, or discard before the next assignment?
 - Now that I've figured out how to pull more data, I probably will add another feature to see if we can get even more improvement from 0.75.
 - What would you try next if data or resources were not constrained?
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8. Mismatch Acknowledgment (Complete Only If Applicable) -

Not applicable for this project at this stage. I think it was a good fit.

If this week's technique was a poor fit, explain:

- Why it does not align with your project:
 - Evidence supporting that conclusion:
 - What value this attempt still provided:
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Submission Notes

- Written submission format: **Markdown or PDF**
- Code or notebooks: **optional unless explicitly requested**
- Performance is **not** graded competitively
- Clear reasoning and honest reflection matter more than results